

Improving Listening Skills Using Artificial Intelligence: A Comprehensive Study

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Abstract—Artificial intelligence (AI) is rapidly transforming language education, especially in the development of listening skills for second-language learners and professional communication training. Recent advances in automatic speech recognition (ASR), natural language processing (NLP), and adaptive learning platforms have enabled intelligent systems that provide real-time feedback, personalized listening tasks, and rich analytics on learner performance. Empirical studies report that AI-enhanced instruction can significantly improve listening comprehension scores, increase learner motivation, and promote self-regulated learning compared with traditional methods. This paper examines the role of AI in improving listening skills by reviewing current technologies, instructional designs, and empirical evidence, with a focus on ASR-based feedback, smart voice assistants, and AI-driven listening platforms. A conceptual methodology is proposed for an AI-powered listening tutor that integrates ASR, NLP-based comprehension analysis, and an adaptive recommendation engine to generate individualized listening tasks and feedback. The discussion highlights strengths such as personalization, scalability, and objective assessment, as well as weaknesses including accent bias, technical constraints, and overreliance on automated feedback. Finally, the paper outlines future research directions, including multimodal listening analytics, affect-aware tutoring, and deeper integration of AI tools into blended classroom pedagogy.

Index Terms—Artificial Intelligence, Listening Skills, Automatic Speech Recognition, Language Learning, Educational Technology

I. INTRODUCTION

Listening is a foundational component of communicative competence and plays a critical role in language acquisition, academic success, and professional interaction [15], [17]. It underpins learners' ability to follow classroom instruction, participate in discussions, and engage with authentic media, yet it often develops more slowly than other skills such as reading or grammar [17], [26]. For many learners of English and other second languages, listening comprehension is particularly challenging due to fast speech, varied accents, connected speech phenomena, unfamiliar vocabulary, and limited exposure to authentic audio input outside the classroom [13], [14]. These difficulties are often intensified in exam-oriented systems where listening is tested but not systematically trained, leading to anxiety and low confidence among learners [13], [16].

Traditional listening pedagogy often relies on fixed audio materials, delayed teacher feedback, and one-size-fits-all tasks that do not adequately reflect individual differences

in proficiency, pace, or learning style [17], [18]. In many classrooms, listening activities follow a predictable pattern of “play audio—answer questions—check answers,” with limited opportunities for interactive practice, repeated exposure, or personalized scaffolding [11], [26]. Teachers frequently face large class sizes and time constraints, making it difficult to provide individualized feedback on comprehension problems or to address diverse listening needs, such as specific accent exposure or training in bottom-up versus top-down listening processes [11], [17]. As a result, learners may adopt test-taking strategies rather than developing robust, transferable listening skills.

AI technologies offer a new paradigm for listening instruction by enabling systems that automatically recognize speech, analyze learner responses, and adapt content in real time [4], [6], [7]. Modern ASR systems based on deep neural networks and transformers can transcribe speech with high accuracy across diverse acoustic environments, providing the foundation for listening comprehension tasks with immediate feedback on both content and pronunciation [1], [3], [5]. Speech analytics and educational data mining techniques can extract additional information from learner interactions—such as error patterns, timing, and persistence—supporting fine-grained diagnosis and adaptive support [8], [9], [31]. AI-driven language learning platforms and smart voice assistants can deliver interactive listening activities, simulate conversational scenarios, and track learner progress with dashboards that visualize development over time [11], [21], [22].

Recent empirical work indicates that AI-enhanced instruction can significantly improve listening comprehension and related skills. Studies report higher post-test listening scores for experimental groups using AI tools compared with control groups using conventional methods, with gains linked to features such as adaptive task difficulty, instant feedback, and rich exposure to varied audio [13], [14], [16]. AI platforms deployed in primary, secondary, and higher education have been shown to support listening, speaking, and pronunciation simultaneously, contributing to improved overall oral proficiency and learner confidence [15], [27], [29]. Meta-analytic evidence further suggests that AI-based instruction has a large positive effect on speaking and listening outcomes, while also enhancing motivation and self-regulated learning through self-paced, interactive environments [11], [12]. At the same time, research highlights persistent challenges such as accent

recognition errors, limited depth of automated feedback on higher-order comprehension, and digital access gaps, all of which underscore the need for careful pedagogical design and responsible implementation [13], [18], [19].

Within this context, there is a clear need to synthesize current knowledge on how AI can best support listening development and to outline practical architectures that can guide future systems. Existing studies investigate diverse tools—ASR-based pronunciation trainers, AI chatbots, smart voice assistants, and integrated learning platforms—but they are often reported in isolation, making it difficult for educators to see the overall pedagogical landscape [28], [32], [33]. Moreover, many implementations focus on specific sub-skills (such as pronunciation or test practice) rather than offering an integrated approach that combines listening comprehension, pronunciation awareness, and metacognitive strategy support in a unified environment [26], [29], [30]. There is also limited discussion of how AI-generated analytics can be meaningfully interpreted by teachers and learners to inform classroom decisions and self-regulated learning pathways [23], [31].

The aim of this paper is to analyze the role of AI in improving listening skills by synthesizing existing literature, outlining a conceptual AI-based listening tutor, and critically discussing strengths, weaknesses, and future directions. Building on recent empirical and theoretical work, this study positions AI not as a replacement for human instruction but as an intelligent partner that can provide scalable, personalized, and data-rich listening support [11], [12], [26]. The key contributions are:

- A structured literature review of AI applications for listening instruction, covering ASR-based tools, AI chatbots, smart voice assistants, and integrated learning platforms;
- A proposed methodology for an AI-powered listening skills platform integrating ASR, NLP, speech analytics, and adaptive learning mechanisms grounded in empirical findings;
- A pedagogically oriented discussion of design considerations, strengths, limitations, and ethical issues, offering recommendations for researchers, system designers, and educators who wish to integrate AI into listening curricula.

II. LITERATURE REVIEW

AI for listening instruction builds on advances in ASR, speech analytics, and intelligent tutoring systems. ASR pipelines convert spoken audio into text through acoustic modeling, language modeling, and decoding, increasingly using deep learning and transformer-based architectures [2], [4], [6]. Transformer-based ASR systems have achieved substantial reductions in word error rate compared with earlier RNN and HMM-DNN baselines, even in noisy environments, enabling more reliable use of recognition outputs in learning applications [1], [3], [5]. Recent surveys highlight how transformer-based ASR models are increasingly applied in educational and call center domains, where robustness to accent and environment is essential for feedback quality and analytics [4], [8],

TABLE I: Representative AI tools and their contributions to listening development

Study / Tool	AI Technology	Main Listening Contribution	Level
WJEL (2024) [13]	ASR-based listening platform	Improved EFL listening comprehension via contextualized tasks and instant feedback	Univ.
IJiet (2025) [14]	SUNO.AI TTS + adaptive tasks	Enhanced Arabic listening scores through personalized exercises and varied dialect exposure	Univ.
HRMARS (2025) [15]	AI learning platform	Better listening and speaking performance in ESL primary learners using interactive tasks	Primary
ESI Culture (2024) [26]	Mixed AI-driven media	More engaging, learner-centred listening practice in EFL classrooms	Secondary
EDUPIJ (2025) [22]	Smart voice assistant	Significant gains in listening and speaking using voice-based practice scenarios	Univ.
ReCALL (2022) [32]	AI chatbot partner	Increased exposure to spontaneous listening and speaking in conversational settings	Univ.

[10]. Speech analytics platforms further analyze transcribed speech to extract features such as keywords, discourse patterns, sentiment, and paralinguistic cues, supporting detailed feedback and performance monitoring in training and learning contexts [8]–[10].

Empirical studies have begun to evaluate AI’s effect on listening skills specifically. A study on AI-assisted listening comprehension in Saudi EFL learners reported significant improvements in post-test scores and enhanced engagement when learners used AI tools offering contextualized listening tasks and feedback on accuracy, fluency, and comprehension [13]. A randomized controlled trial on AI-driven speech recognition in EFL listening showed that the experimental group using ASR-based tools achieved higher listening comprehension, stronger flow experience, and reduced listening anxiety than the control group [24], [25]. Research integrating AI platforms such as SUNO.AI for Arabic listening found notable gains in post-test listening scores and emphasized the role of personalized exercises and exposure to varied dialects in improving listening performance [14]. Work on AI-supported listening and speaking in ESL primary learners similarly showed that AI tool groups outperformed basic groups in overall performance, underscoring the impact of interactive, adaptive platforms on young learners’ listening and speaking skills [15], [27].

A growing body of research explores ASR-based systems as a foundation for listening and pronunciation training. ASR-based pronunciation programs such as BoldVoice and other mobile-supported tools have been found to significantly improve learners’ fluency, segmental accuracy, and overall communicative competence when compared with conventional

classroom instruction, highlighting the value of infinite, on-demand ASR access for practice [28]–[30]. Studies with Korean and other EFL learners show that guided ASR practice leads to measurable gains in accentedness, comprehensibility, and global speaking skill, and that learners perceive ASR feedback as beneficial for pronunciation and listening awareness [20], [29], [30]. Monitoring of learner behaviour in autonomous ASR-based practice further reveals how learners distribute effort across sessions and how usage patterns relate to gains, providing design insights for listening and pronunciation tasks [31].

Broader meta-analyses and mixed-methods studies in AI-mediated language instruction provide additional context for listening development. A meta-analysis on AI in English instruction reported a large positive effect size for AI-enhanced speaking and listening, emphasizing the benefits of personalized, interactive environments built on speech recognition, conversational agents, and adaptive platforms [12]. Another mixed-methods study on AI-mediated English learning found that AI-based platforms improved learning outcomes, motivation, and self-regulation compared with traditional instruction, recommending blended integration rather than standalone use [11]. A theoretical review on using AI-driven tools for teaching listening skills concludes that AI-aided media can make listening practice more engaging and learner-centered, while synthesizing a wide range of classroom implementations and outcomes [17], [26].

Conversational agents, voice chatbots, and smart voice assistants form another important strand of literature relevant to listening improvement. A study implementing an AI voice chatbot as an English conversation partner in EFL classrooms reported high turn-taking rates and increased willingness to communicate, suggesting that such systems can provide rich listening and speaking exposure rarely achieved in conventional classes [32]. Research on AI chatbots and platforms in higher education shows that learners use chatbots to prepare for listening, speaking, reading, and writing tasks and perceive them as useful in building confidence and fluency [33]. Smart voice assistant systems designed specifically to develop listening and speaking skills have also demonstrated significant improvements in learners’ performance and engagement, supporting the integration of modern AI voice technologies into language curricula [21], [22].

At the same time, literature cautions about limitations in current AI tools for listening. Researchers note issues related to accent recognition, instability under noisy conditions, limited pragmatic or discourse-level feedback, and potential overreliance on automated evaluation metrics [5], [13], [19], [20]. Several empirical studies report algorithmic and accuracy constraints in ASR, especially for non-native speakers, and highlight the risk of learners receiving misleading feedback if recognition errors are not handled carefully [27], [29], [30]. Concerns about data privacy, transparency of AI algorithms, and the authenticity of synthetic voices also arise in studies analyzing learner perceptions and ethical implications [16], [18], [23]. Overall, the literature suggests that AI can substan-

tially enhance listening instruction when integrated with sound pedagogy and human guidance, but it cannot fully replace teacher mediation or rich classroom interaction, especially for higher-order listening skills and socio-cultural aspects of communication [11], [12], [26].

III. METHODOLOGY

This section proposes a conceptual AI-powered listening skills tutor that leverages ASR, NLP, and adaptive learning to deliver personalized listening practice and feedback. The system is designed as a modular architecture with five core components: audio delivery and capture, ASR and speech processing, comprehension and pronunciation analysis, adaptive task recommendation, and analytics and teacher dashboard.

A. System Overview

Fig. 1 illustrates the overall workflow. Learners interact with listening tasks through a web or mobile interface, while the back-end services handle speech recognition, analysis, and personalization.

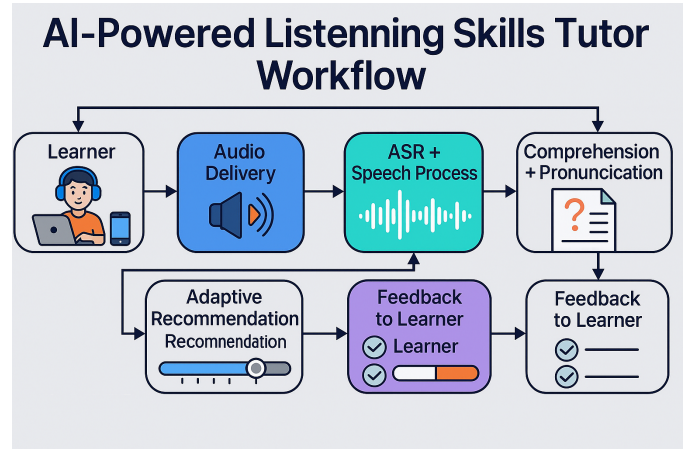


Fig. 1: Conceptual workflow of an AI-powered listening skills tutor.

B. Audio Delivery and Learner Interaction

Learners access listening tasks such as dialogues, lectures, announcements, or role-plays through an intuitive interface. Tasks include multiple-choice comprehension questions, open-ended oral responses, shadowing exercises, and dictation-style activities. Audio can be sourced from authentic materials or generated with high-quality text-to-speech, with controls for playback speed, noise level, and accent selection.

C. ASR and Speech Processing

During tasks requiring oral responses, learner speech is captured via microphone and processed by an ASR module built on transformer-based or hybrid CTC-transformer models. The ASR outputs text transcriptions and timing information while handling variations in accents and speaking rates. Domain-specific fine-tuning and data augmentation strategies are adopted to improve robustness in educational settings.

D. Comprehension and Pronunciation Analysis

The system compares ASR output with reference answers or scripts. NLP-based similarity measures assess whether key content words and paraphrases are present, producing a comprehension score. Pronunciation analysis uses acoustic features and phoneme-level alignment to estimate segmental accuracy, stress, and rhythm, summarizing results into learner-friendly feedback such as highlighted mispronounced words or missing information.

E. Adaptive Task Recommendation

An adaptive engine maintains learner profiles and performance histories. It dynamically selects the next listening task by adjusting variables such as audio speed, accent, background noise, and task type. Machine learning models can be trained to predict which task will maximize learning gain based on past responses, error patterns, and engagement data.

F. Analytics and Teacher Dashboard

A dashboard aggregates metrics including listening accuracy by topic, error categories, time-on-task, and progression over sessions. Teachers can inspect class-level and individual data to identify learners needing support, design interventions, and integrate AI practice into blended courses. Researchers can analyze anonymized logs to refine models and instructional designs.

IV. DISCUSSION

AI offers several advantages for listening instruction. First, AI tutors enable fine-grained personalization, adapting task difficulty, audio characteristics, and feedback to individual learner profiles, which is difficult in large classes. Second, ASR-based systems provide immediate and objective feedback on both comprehension and pronunciation, supporting intensive practice cycles. Third, AI platforms extend access to diverse listening materials and contexts beyond textbook audio, including multiple accents and domains.

Nonetheless, challenges remain. ASR errors, particularly for non-native accents and noisy environments, can undermine learner trust and distort feedback. Many systems still focus primarily on surface-level accuracy rather than higher-order skills such as inference, pragmatic understanding, and discourse tracking. Technical and access constraints, especially in low-resource settings, may limit the equitable use of AI listening tools. Ethical concerns about data privacy, algorithmic bias, and the pedagogical consequences of overreliance on automated feedback must also be addressed.

AI listening tools are therefore best deployed within blended learning models. Teachers can use analytics to inform instruction, facilitate metacognitive reflection on listening strategies, and provide human feedback on nuanced aspects that AI cannot yet evaluate reliably. Clear guidelines and training for educators are necessary to ensure that AI augments rather than replaces human interaction.

AI Tools Show Higher Listening Score Gains Across Studies

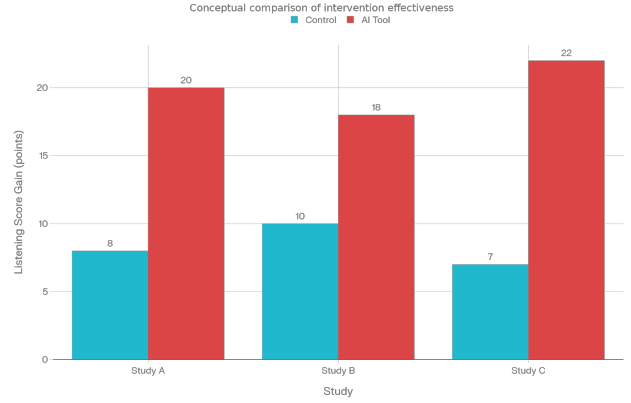


Fig. 2: Conceptual comparison of listening score gains for control and AI-tool groups across three example studies.

V. STRENGTHS

Key strengths of AI for listening skills include:

- **Personalization and adaptivity:** AI-driven platforms can tailor listening tasks, audio complexity, and feedback to each learner's proficiency, goals, and progress, leading to more efficient practice and higher achievement.
- **Immediate feedback:** ASR-based systems offer real-time information about comprehension and pronunciation, enabling learners to engage in intensive, low-stakes practice with rapid cycles of performance and correction.
- **Data-driven insights:** AI systems collect detailed performance data over time, supporting fine-grained analysis of listening strengths and weaknesses for both learners and teachers, and facilitating evidence-based instructional decisions.
- **Scalability:** AI tools can serve large numbers of learners asynchronously, providing listening practice outside classroom hours and in contexts where qualified language teachers are scarce.

VI. WEAKNESSES

Notable weaknesses and challenges include:

- **ASR accuracy and bias:** Recognition errors increase for non-native accents, code-switching, or noisy environments, potentially undermining learner confidence and feedback reliability.
- **Limited higher-order feedback:** Current AI tools are better at assessing surface-level correctness than deep comprehension, pragmatic interpretation, or discourse-level listening strategies.
- **Dependence on technology:** Effective use requires stable internet connectivity, suitable devices, and a degree of digital literacy, which may disadvantage learners in low-resource settings.
- **Pedagogical and ethical concerns:** Overreliance on automated tools may reduce human interaction, and un-

resolved issues around data privacy, transparency, and responsible use of learner data pose risks.

VII. FUTURE SCOPE

Future research on AI for listening skills can advance along several directions. Multimodal analytics that combine audio features, eye-tracking, and interaction logs could yield deeper insights into listening processes, enabling systems to detect confusion, mind wandering, or cognitive load in real time. Affect-aware tutoring, using sentiment and emotion recognition from voice, could help AI systems adjust task difficulty, feedback tone, and pacing to maintain motivation and reduce anxiety.

Technically, transformer-based ASR models fine-tuned on diverse L2 learner corpora and regional accents can reduce bias and improve recognition accuracy for underrepresented groups. Integration of large language models with ASR could support more nuanced, conversational feedback on comprehension and listening strategies rather than simple right-wrong scoring. Pedagogically, stronger frameworks are needed for blending AI listening tools with curriculum design, teacher professional development, and assessment policies, ensuring alignment with learning outcomes and ethical guidelines. Longitudinal studies should also investigate the long-term impact of AI interventions on listening proficiency, learner autonomy, and transfer to real-world communication contexts.

VIII. CONCLUSION

AI is reshaping the landscape of listening instruction by combining advanced ASR, NLP, and adaptive learning techniques to deliver personalized, feedback-rich listening practice at scale. Empirical studies across different languages and educational levels demonstrate that AI-based tools can significantly improve listening comprehension, enhance engagement, and support self-regulated learning compared with traditional approaches. A conceptual AI listening tutor architecture illustrates how learner audio can be captured, automatically analyzed, and used to drive individualized task selection and analytics for both learners and teachers.

At the same time, limitations related to recognition accuracy, depth of feedback, technological constraints, and ethical considerations show that AI should complement rather than replace human instruction. The most promising direction is a blended model in which AI provides intensive, adaptive listening practice and objective data, while teachers orchestrate learning experiences, interpret analytics, and address socio-affective and cultural dimensions. Ongoing research in multimodal, affect-aware, and ethically grounded AI systems can further strengthen the contribution of AI to listening skills development in inclusive and sustainable ways.

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